

Interactive tutorial sheet 8

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Numerical methods

This live script will show you how to use MATLAB for numerical methods such as integration, differentiation. One way uses a vector approach and the other uses the symbolic Toolbox.

Polynomial

This method enters the polynomials as a vector. You have come across this form before when fitting data and the way it produces the value of m and c for $y=mx+c$.

It requires the coefficients to be entered in decreasing order, with zeros for those that are missed out.

For example for the function

$$f(x) = x^3 + x + 3$$

```
clear, clc
p=[3 0 1 3]; % 3 units, one x terms, no x squared terms, 3 cubic terms.
```

The polynomial can then be evaluated for $x=2$ by using polyval

```
a=polyval(p,2)
```

You can then combine this output to plot out the functional form of polynomials such as

```
x=linspace(-5,5,100);  
p=[1 -1 2];           % x^2 - x + 2  
y=polyval(p,x);  
plot(x,y);
```

Integration and differentiation of polynomials

The polynomial can be differentiated/integrated using the functions *polyder/polyint* which return a vector that specifies the coefficients.

Here we differentiate and integrate the following polynomial.

$$y = x^2 - 2x - 3$$

```
clear, clc  
p=[1 -2 -3] % y = x^2 - 2x -3
```

```
p = 1x3  
    1    -2    -3
```

```
polyder(p) % gives 2 -2 meaning dy/dx = 2x -2
```

```
ans = 1x2  
     2    -2
```

```
polyint(p) % Note that it gives the constant as 0
```

```
ans = 1x4    Columns 3:4  
    -3     0
```

Compare your results with you doing them by hand.

TASK 1

Differentiate and integrate the following

- $y = 3x^2$
- $f(x) = 6x^3 - 9x + 4$
- $y = 2t^3 - 10t^2 + 13t$
- $f(x) = 7x^6 - 10$

```
% differentiate
```

```
% integraton
```

TASK 2

Solve the following definite integral

$$\int_1^2 x^2 dx$$

Symbolic toolkit

As you can see the polynominal way is a bit awkward and limited. The Symbolic Math Toolbox lets you analytically perform differentiation, integration, simplification, transforms, and equation solving in a much easier way.

While it is available on the online version, if you have installed this yourself locally you will need to install the toolbox. It can be searched for under Add-Ons.

A good link is here. <https://uk.mathworks.com/matlabcentral/answers/101885-how-do-i-install-additional-toolboxes-into-an-existing-installation-of-matlab>

Symbolic variables

Symbolic values are exact representations, unlike floating-point numbers (such as what we have been used previously). To generate them you need to specify a symbolic variable using the key term `sym`. Without this it will not work as a symbolic value but a just a standard variable.

When you use the key term **`syms`**, it creates a new symbolic variable that does not have any assumptions on how it will be used.

```
clear, clc
syms x1 % set up variable
x1      % Note that this gives x as an answer.
```

```
x1 = x1
```

If you now look in the workspace. This generates a symbolic variable x as denoted by 1x1 sym. Before you continue, make sure this works, as if it does not then the symbolic toolbox is not installed correctly.

Symbolic numbers

To show the difference, we can create a symbolic number by using *syms* and compare it to the same floating-point number.

As we are dealing with just values in this case, we need to use the **sym** a little differently.

```
syms x;
x=sym(1/3);           % symbolic variable
y=1/3;                % double variable

x                     % exact fraction
y                     % double

x=x*2                 % exact fraction
y=y*2                 % double

fprintf("%s\n", x);   % have to use %s to display a symbolic variable
z=double(x*2)         % Convert it to a double
fprintf("%f\n", z);   % now converted can use %f to display
```

Note how x remains as a fraction (exact) and the floating point effectively rounds the value due to the structure of the floating point variable (also note that the font it uses is slightly different!)

Symbolic equations

Using the symbolic equations you can solve and rearrange formula very quickly.

Let us try to form the equation $y = 3x^2 - 2x$. To do this you must

- Create two symbolic variables, x and y . Both will appear in your workspace as a 1x1 sym.
- We shall set the answer to a third variable, here called *eqn*. *eqn* is just a variable but equal (=) to the equation we are wanting to solve.
- To avoid confusion in writing the equation and defining variables, y is said to be equivalent (==) to $3x^2 - 2x$, where the *eqn* is equal (=) to the defined symbolic equation.
- *eqn* does not need to be defined as symbolic variable as it inherits that from the symbolic variables x and y . So if you check the workspace you will see that *eqn* is also a 1x1 sym.

```
clear, clc
syms x y               % define a symbolic variable for x and y
eqn=y==3*x^2-2*x;      % defines the equation assigned to a variable of eqn
fprintf("%s\n", eqn);  % note this holds full equation in symbolic form
```

Once set up you can then work with these symbolic equations to rearrange and solve your equations.

```
a=solve(eqn,y) % rearrange equation for y (no change)
b=solve(eqn,x) % rearrange equation for x - note two solutions.
c=subs(eqn,x,3) % substitute x for the value of 3
```

Working with symbolic equations

You can also factorise, expand and simplify equations. These are somewhat limited but a good starting point for many engineering problems you will encounter.

```
clear, clc
syms x % set up symbolic variable.
eqn = x^2-1 % set up equation here x^2-1.
a=factor(eqn) % factorise the equation.
eqn = (x+5)*(x-2) % new equation
b=expand(eqn) % expand the equation.

eqn=(1 - x^2)/(1 - x)
c=simplify(eqn) % simplify the equation

eqn=(x - 1)*(x + 1)*(x^2 + x + 1)*(x^2 + 1)*(x^2 - x + 1)*(x^4 - x^2 + 1)
d=simplify(eqn) % Can be very useful for complex equations.
```

You can rearrange complex functions in exactly the same way.

```
clear, clc
syms P V n R T
eqn=P*V==n*R*T
a=solve(eqn,V) % rearrange for V
b=solve(eqn,P) % rearrange for P
c=subs(eqn,P,2) % substitute P for the value of 2
```

Note that you can only subs a value into one symbolic variable at a time with subs. To do more you need to do it one by one, equalling the new equation to another variable as you go along.

TASK 3

Set up the equation of motion $s=0.5at^2$.

Set acceleration, a , to be 9.81 ms^{-1} and using the symbolic variables, solve for when $t=2\text{s}$.

```
% create symbolic variables.

% define equation - set to a variable - for example eqn
```

```
% Use equation variable and subs to set a=9.81. Set answer to another variable e.g.
eqn2
```

```
% Now subs the value of t=2 into this equation.
```

Plotting symbolic functions

You can plot functions using the `fplot` command. It requires the function (equation in symbolic form). You can also define limits (here between -6 and 4), points, lines colour etc much like `plot`.

```
clear, clc
syms x
eqn=x^2-4;           % set up the equation
fplot(eqn,[-6 4], 'or') % plot the equation
solve(eqn)           % at y=0 x is either 2 or -2
```

TASK 4

Plot $\tan(x)$ between -10 and 5 using `fplot`.

Modify your code to show a red cross at the point of $x=-5$.

TASK 5

Set-up and rearrange the Brinell equation in terms of d . You can find the Brinell equation in your MAT1643 lecture notes.

Think about your answer and what it means.

Modify your code such that for given (user input) values of HB , D and F , it will calculate the value of d .

Simultaneous equations

We solved an equation using the `solve` function, but we can use `solve` on a set of equations together.

For instance, If we had the following set of equations

- $2x+y+z=2$
- $4-x+y=3z$
- $x+2-2y=z$

We can solve them simultaneously. Note we need to inform MATLAB that there are 3 outputs, be careful of the order!

```
syms x y z
eqn1=2*x+y+z==2;
eqn2=4-x+y==3*z;
eqn3=x+2-2*y==z;
[x, y, z]=solve(eqn1,eqn2,eqn3); %solve simultaneously with 3 outputs
fprintf("x=%s, y=%s, z=%s", x,y,z);
fprintf("x=%.2f, y=%.2f, z=%.2f", double(x),double(y),double(z));
```

TASK 6

Using the symbolic toolkit to solve the following set of simultaneous equations.

$$\begin{aligned} 3z+6x &= 2 \\ -6x+y &= 6z \\ -3y+6 &= -z \end{aligned}$$

TASK 7

A student has answered the following question but has one mistake in their code which has led them to getting the wrong solution.

Question: Solve the following simultaneous equations

$$x+6y=2$$

$$2x-3y=0$$

The code they have produced is below. Can you correct it for them?

```
syms x y
eqn1=x+6*y-2==0;
eqn2=2*x-3*y==0;
[a, b]=solve(eqn1,eqn1);
fprintf("x=%.2f, y=%.2f\n", double(b),double(a));
```

Integration and differentiation

One particular strength of the symbolic toolkit is to be able, in one line, to integrate and differentiate your symbolic functions.

```
clear, clc
syms x

a=diff(cos(x)) % integrating cos(x)
b=int(sin(x))  % differentiating sin(x)

c=diff(log(x)) % differentiating ln(x)
d=int(1/x)     % integrating 1/x remember log is ln in matlab
```

You can also use this with symbolic variables and equations

```
clear, clc
syms x
eqn=2*x+1
a=diff(eqn) % differentiate x
b=int(eqn)  % integrate x
b2=expand(b) % multiply out the integration
c=int(eqn,1,2) % integrate x between limits of 1 and 2
d=eval(c) % Evaluate the fractional form to a value
```

Here we plot the function and the differential(s) of the function

```
clear, clc
syms x
eqn=x^2 % sets up a symbolic variable holding x^2
diff_eqn=diff(eqn) % can then differentiate that wrt x

fplot(eqn,[-6 4],'-r');
hold on
fplot(diff_eqn,[-6 4],'-b')
fplot(diff(diff_eqn), [-6 4],'-g')
hold off
```

Multivariate fuctions

These are functions with more than one variable. They have to defined a little differently.

```
syms x z
f(x,z) = x/(1+z^2) % define equation based on two variables
```



```
dx=diff(f,x)

Fx=int(f,x)      % integrate with respect to x
Fy=int(f,z)      % integrate with respect to y
```

If you do not specify the variable to integrate against, it will pick the first one.

TASK 8

In mathematics, a parametric equation defines a group of quantities as functions of one or more independent variables called parameters.

A circle of radius 1 can be expressed as

- $x=\cos(t)$
- $y=\sin(t)$

First use this to plot a circle using the symbolic toolkit and fplot.

Modify your code to generate a circle with radius of 2 with a centre located at $x=1$ and $y=0$

TASK 9

Below is a set of data for the Young's modulus of a metal with a variable amount of additive.

Additive (g) Young's Modulus (GPa)

0.0	125
0.3	139
0.4	142
0.6	146
0.7	145
0.9	143
1.1	138

Plot out the data. There appears to be a potential maximum in the Young's modulus.

Fit the data with a polynomial. Use this formula to estimate the amount of additive that leads to the maximum Young's modulus (to 2 dp).

```
clear, clc

% plot the data
```

```
% fit the data
```

```
% using the fit equation, differentiate it and solve for the turning point (i.e.  
when equal to zero)
```

```
% print answer to the screen
```

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